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Self-Consistent Modelling Strategies for Microwave Plasma Power Deposition across Multiple Regimes

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Abstract

Sample text inserted for demonstration. Replace with abstract text. Your abstract must give readers a brief summary of your article. Concisely describe the contents of your article and include key terms. It should be informative and accessible: indicate the general scope of the article and state the main results obtained and conclusions drawn. The abstract must be complete in itself: it must not contain undefined abbreviations and must not refer to any table, figure, reference or equation numbers. Normally the abstract text is not more than 300 words.

1 Introduction

1.0.1 1st and 2nd Paragraphs: Microwave plasma technologies have been shown to be competitive in the realm of chemical conversion, namely for nitrogen fixation [1–4]. Nonetheless, working parameters still need to be optimized to facilitate the transition from traditional methods of chemical conversion to this electrified approach. Microwave plasmas can be operated in a large variate of pressure and power conditions resulting in different plasma parameters suitable for specific applications [5–7].

There is also strong scientific and engineering interest in understanding the spatial structure of microwave plasmas, which can adopt a wide variety of shapes depending on operating conditions [8–10]. The plasma shape results from the balance between power absorption and its dissipation through radial conduction and diffusion of energetic particles, as these processes significantly affect each other. From the engineering standpoint, the shape of the plasma is critical because it defines the volume where chemical conversion occurs and strongly influences gas flow behavior [4, 11]. Because the microwave field distribution itself depends on the evolving plasma properties, predicting this equilibrium is intrinsically a coupled electromagnetic–plasma problem.

1.0.2 Objectives of work: The primary objective of this study is to develop a self-consistent model that couples microwave propagation with plasma fluid equations. The model computes the absorbed power density, which is a key quantity controlling electron density, electron temperature, and chemical kinetics in microwave plasmas. A secondary objective is to make this model sufficiently flexible so that it can be coupled to different types of plasma fluid models from 1D to multidimensional formulations.

1.0.3 Justification for these objectives: Most existing plasma models rely on predefined power deposition profiles, which are typically fitted to experimental emission or based on simplified analytical assumptions [10–15]. However, in microwave plasmas, electron density couples strongly to the local electric field meaning that changes in plasma parameters can significantly modify the electromagnetic wave propagation, making it essential to have a self-consistent approach [10, 12, 16]. This component of plasma modelling is specially important because of the difficulty to experimentally probe the electric field distributions in microwave plasmas. This way, models allow to fill the gaps in experimental results and to predict the behavior of untested geometries and parameter spaces.

1.0.4 Background: What else has been done before? How ? What have WE done previously?

Fluid models, namely 1D fluid models have been widely used in microwave plasma modelling with relatively good agreement with experimental data [10, 12, 17]. On the other hand, the profiles used for the absorbed power density are predefined profiles in the vast majority of simulations. Furthermore, sensitivity analyses on fluid models show that species densities and temperatures can vary substantially depending on the assumed power profile [12, 13]. This study builds on a previously validated 1D plasma fluid model [10, 12], which until now used predefined power profiles. This work extends this model by coupling it with a microwave electromagnetic solver, obtaining the power deposition self-consistently. It should be noted that in recent years there have been a few specialized models that included self-consistent field calculations [18–22].

1.0.5 Guidance to reader: What strategy used? What highlights from this paper? What to look for? What can the reader expect as a conclusion? This study introduces a new electromagnetic model to calculate the electric field amplitude and the corresponding absorbed power density in realistic microwave plasma reactors. The model is applied in two complementary ways:

- A 2D electromagnetic simulation coupled with a 1D plasma fluid model, enabling a self-consistent evaluation of power deposition inside the plasma tube;
- A 3D electromagnetic simulation of the same reactor extended to include the 3-stub tuner to evaluate the validity of the power normalization assumption done in the 2D case.

The geometry of the waveguide system is highlighted as an important component of the modelling of the microwaves and of the microwave plasma as a whole.

A key result of this work is the identification of distinct microwave propagation regimes based on conductivity of the plasma. These regimes help us determine when simplified electromagnetic models are sufficient and when the full Maxwell system needs to be properly solved self-consistently. Under this proposition, simpler models are tested in each regimes understanding the accuracy and numerical cost of each.

1.0.6 Summary: What are the sections of the paper? The paper is organized as follows. In section 2, the geometry of the system being studied is introduced. Section 3 presents the theoretical framework, including governing equations, boundary conditions for both electromagnetic and fluid models, and the numerical implementation details. Section 4 presents the simulation results, including self-consistent plasma simulations in 1D-2D, microwave absorption and propagation analysis, 3D waveguide modeling with impedance matching and comparison of electric field models. Conclusions are drawn in section 5.

2 Geometry of the Reactor

- This section contains the schematic of the reactor together with simplified diagrams explaining each component and it's purpose on the reactor.
- The quantities R_{in} , R_{out} , a , b , y_r and $z_{1,2,3}$ are defined here.
- The WR-340 waveguide, the 3-stub tuner and the MW generator are referenced by the exact dimensions.

Here I need to refer how the simulation models the plasma in the central slice in the XY dimensions. And mention how for higher plasma conductivities the majority of energy being absorbed by the plasma turns into the axial edges of the plasma relative to the direction of the excitation field.

3 Model

3.1 Governing Equations

3.1.1 Electromagnetic Equations

Maxwell's equations in harmonic time

$$\vec{\nabla} \cdot (\epsilon_0 \epsilon_r \vec{E}) = \rho_c, \quad (1)$$

$$\vec{\nabla} \cdot \vec{H} = 0, \quad (2)$$

$$\vec{\nabla} \times \vec{E} = -\mu_0 i \omega \vec{H}, \quad (3)$$

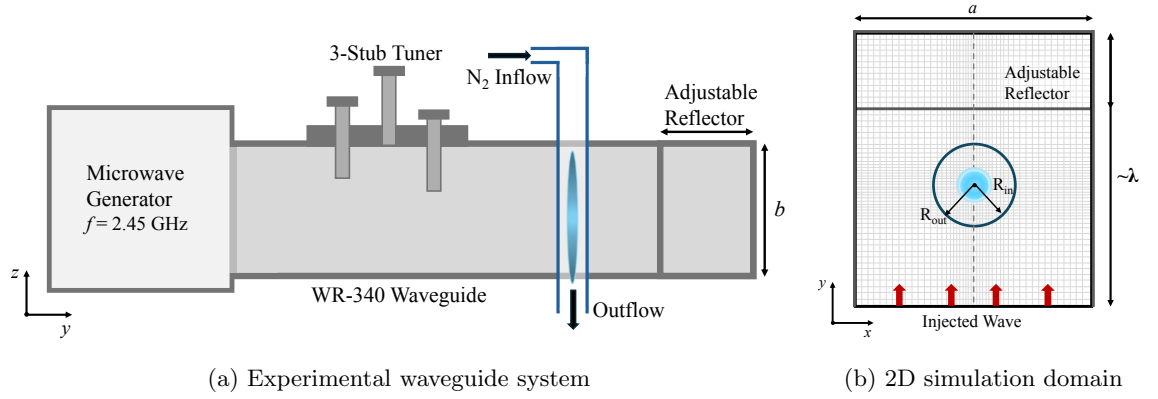


Figure 1: Schematics of the waveguide system ($a = 86.36$ mm, $b = 43.18$ mm) in a YZ cross section showing the full system, (a) and a XY cross section showing the simplified simulation domain, (b). The grid lines are shown only to indicate the non-uniform nature of the computational mesh and do not represent the actual grid used in the simulations.

$$\vec{\nabla} \times \vec{H} = \vec{J} + \epsilon_0 \epsilon_r i \omega \vec{E}, \quad (4)$$

where $\hat{\sigma}$ is the complex conductivity and $\hat{\epsilon}_r$ is the complex permittivity defined as

$$\hat{\epsilon}_r = \epsilon_r - \frac{i \hat{\sigma}}{\omega \epsilon_0}. \quad (5)$$

Final equation being solved

$$\nabla^2 \vec{E} + \vec{\nabla} \left(\frac{\vec{\nabla} \hat{\sigma} \cdot \vec{E}}{i \omega \epsilon_0 \hat{\epsilon}_r} \right) + k_0^2 \hat{\epsilon}_r \vec{E} = 0. \quad (6)$$

Power density from electric field

$$P_{dens} = \frac{1}{2} \text{Re} [\hat{\sigma}] \|\vec{E}\|^2. \quad (7)$$

3.1.2 Fluid Equations Fluid model comes from Altin 2024 [12] which is based on Vialleto2022 [10]. The equations are bellow

Mass fraction

$$\frac{\partial(\rho y_i)}{\partial t} = -\frac{1}{r} \frac{\partial[r(J_i)_r]}{\partial r} + m_i S_i + \Omega_i. \quad (8)$$

Temperature

$$\frac{\partial T_g}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[\alpha r \frac{\partial T_g}{\partial r} \right] - \sum_i \frac{1}{r} \frac{\partial[r(J_i)_r h_i]}{\partial r} + \frac{Q + Q_{flow}}{\rho c_p}. \quad (9)$$

Electron energy

$$\frac{\partial n_e}{\partial t} = \frac{1}{r} \frac{\partial[r(J_e)_r]}{\partial r} + S_e. \quad (10)$$

- Equations defining flow

- Equations defining source terms (ω_i , Q_{in} and S_e)

- Explanation that power density in 1D model is scaled up to input power because of imperfect impedance matching.

- Further details are pointed to papers of origin of this model

3.2 Boundary conditions

3.2.1 Electromagnetic Model References to Figure 1b explaining exactly which surfaces use which boundary conditions.

1. Dirichlet (Metal Surfaces): $\vec{E}(x) = \vec{E}_{forced}(x)$
2. Neumann (Symmetry Planes): $\frac{\vec{E}_i(x)}{dx_j} = \vec{\phi}(x)$

3. Robin (Wave Injection Boundary): $\frac{\vec{E}_i(x)}{dx_j} - ik\vec{E}_i(x) = -2ik\vec{E}_{injected}(x)$
4. Open Boundaries (Perfectly Matched Layers):

$$s(x) = 1 - \frac{i\sigma_s(x)}{\omega\epsilon_0}, \quad (11)$$

where $\sigma_s(x)$ is chosen to ensure the decaying field is as smooth as possible for the specific geometry and x is the direction of the open boundary. Following Berenger *et al* [23], in this study we use

$$\sigma = \frac{0.8 \times 2\pi(m+1)c\epsilon_0}{n_{\text{PML}}dx}, \quad (12)$$

with the parameter $m = 5$ while n_{PML} represents the number of layers in the PML boundary condition and it is chosen to be between 10 and 20 depending on the problem.

3.2.2 Fluid Model Similar to boundary condition section from Altin2024.

Dirichlet, Neumann and Robin boundary conditions are also available on the fluid model. Neumann boundary conditions are used on axis of symmetry.

On the boundary with the quartz tube the fluxes of the continuous functions calculated are altered to account for that interaction. The thermal flux at the inner wall of the tube is given as [10, 24]

$$J_g(R_{in}) = -\lambda_w \frac{T_g^{env} - T_g(R_{in})}{R_{in} \ln\left(\frac{R_{out}}{R_{in}}\right)}, \quad (13)$$

For the density of species the diffusive flux is substitute for the following term containing the probability, γ , of particles being absorbed in the wall upon collision.

$$J_i(R_{in}) = \frac{\gamma_i}{2 - \gamma_i} \left(\frac{q_i}{|q_i|} \mu_i n_i \vec{E}_{amb} - \frac{1}{2} n_i v_{th} \right), \quad v_{th} = \sqrt{\frac{8k_B T_g}{\pi m_i}} \quad (14)$$

Finally, the electron energy density flux at the boundary is similarly calculated

$$J_e(R_{in}) = \frac{\gamma_e}{2 - \gamma_e} \left(-\mu_e n_e \vec{E}_{amb} - \frac{1}{2} n_e \epsilon v_{th,e} \right), \quad (15)$$

3.3 Chemical Set

- Details on the chemical set used. Reference to original paper using this set. No table with all reactions needed [12, 25, 26].
- Refer that vibrations are considered in STS.

3.4 Electron Kinetics

- References to the original works introducing these electron kinetics sets. Details on important deviations from the original works [25–29].
- Electron kinetics reaction rates calculated from EEDF separately on each radial position
- Solved for the (E/N) value corresponding to the mean energy calculated using equation (10).

3.5 Numerical implementation

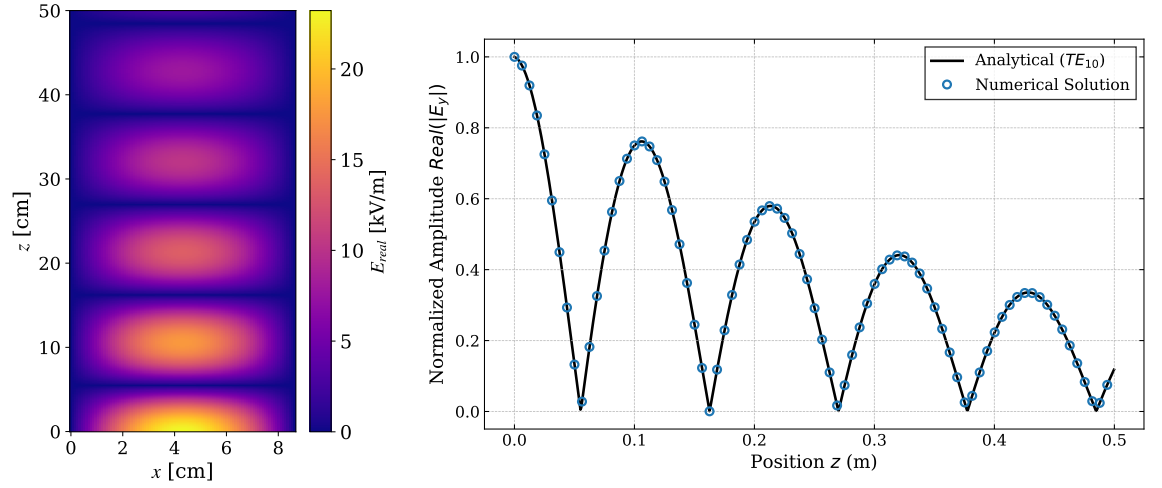
- Details of spatial discretization (figure 1b can contain grid lines and be referenced here)
- Details on time discretization (CFL, on 1D fluid model and on coupling, EM solver is harmonic, also for coupling with BOLSIG+, reflector position update)
- Details on solver algorithms:

EM model: superLU direct solver

1D fluid model: Thomas algorithm

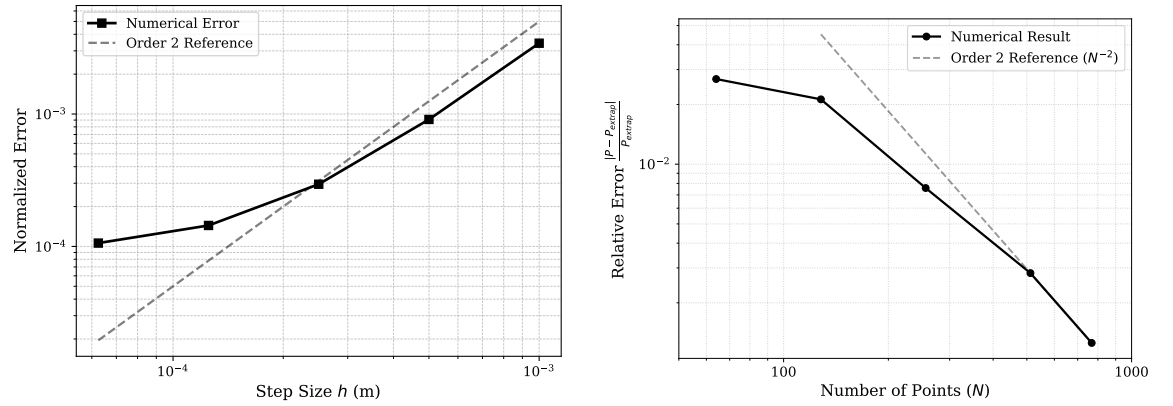
EBE: BOLSIG+

- Computational time details



(a) Numerical solution on XZ cross section (b) Comparison of the normalized amplitude $Real(|E_y|)$ along the central axis ($x = a/2, y = b/2$). Every tenth numerical point shown.

Figure 2: Comparison of numerical result, using 100 points in x direction and 800 in the y direction, with analytical solution for TE_{10} mode inside a WR-340 waveguide filled with uniform plasma ($n_e = 6 \times 10^{16} \text{ m}^{-3}$, $\hat{\mu} = 0.8 - 2.5 \text{ m}^2/(\text{V s})$).



(a) Field amplitude inside WR-340 waveguide relative to the analytical solution.

(b) Absorbed power in 2D setup from 1 relative to the Richardson extrapolation.

Figure 3: Numerical convergence and error analysis.

4 Results

4.1 Verification of the Electromagnetic Solver

4.2 Electric Field and Reflector Tuning

First part of section is introducing field and results getting optimized with the reflector.

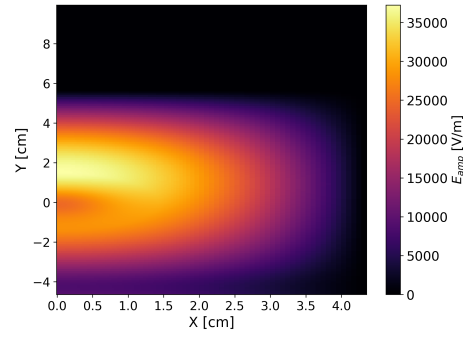


Figure 4: Two-dimensional distribution of the electric field magnitude in the waveguide-plasma system after reflector tuning, for a fixed plasma density profile and injected microwave power of 770 W.

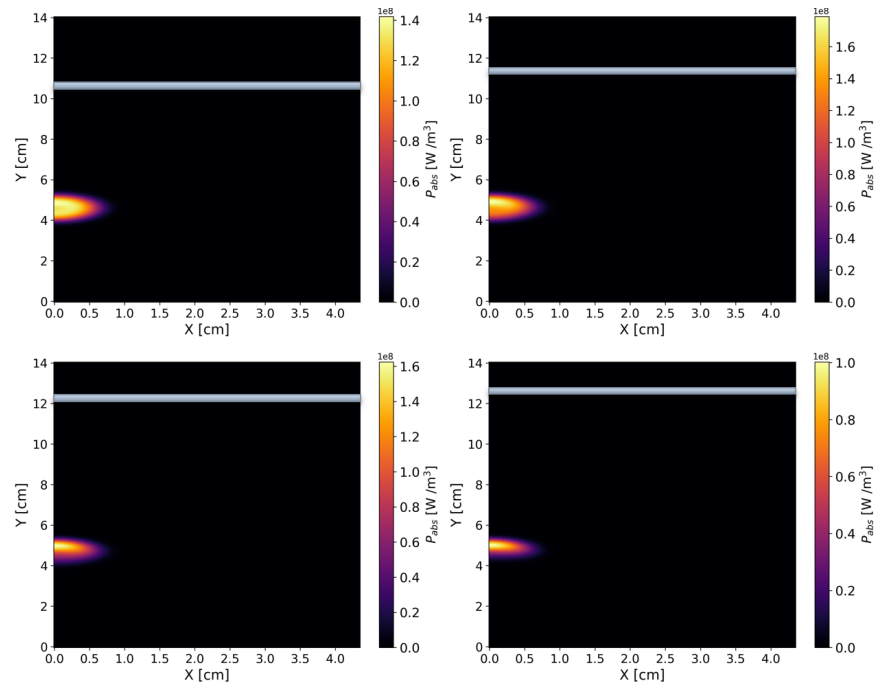


Figure 5: Many graphs showing P_{abs} for different positions of reflector

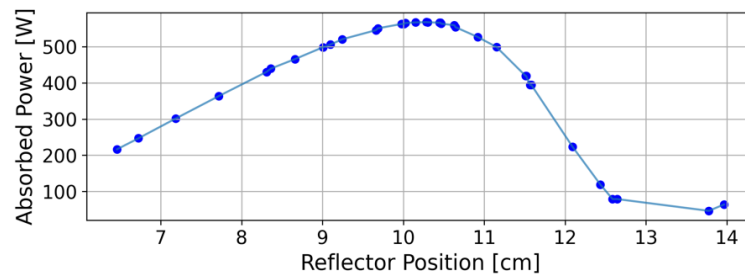


Figure 6: Total absorbed power of a 770 W microwave inside a plasma of (plasma to be specified) as a function of the reflector position.

Here I reference how not all power is absorbed and how this would change in the full 3D system with the possibility of including a figure like this with the tuner being optimized in 3D. I do not go into details about this but instead need to refer that for denser plasmas the assumption of the core region being the most important and representative of the power absorption are no longer valid.

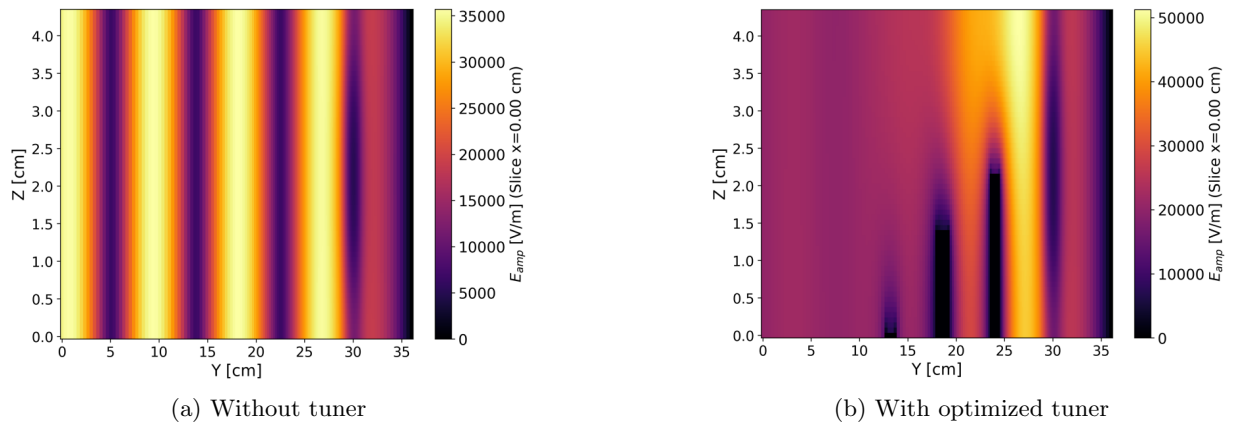


Figure 7: 2D slices of electric field magnitude inside the waveguide from 3D simulations. (not final)

4.3 Self-consistent Plasma Simulation

Simulation in nitrogen(at least) showing how deposited power and all other plasma quantities change with pressure and input power and flow.

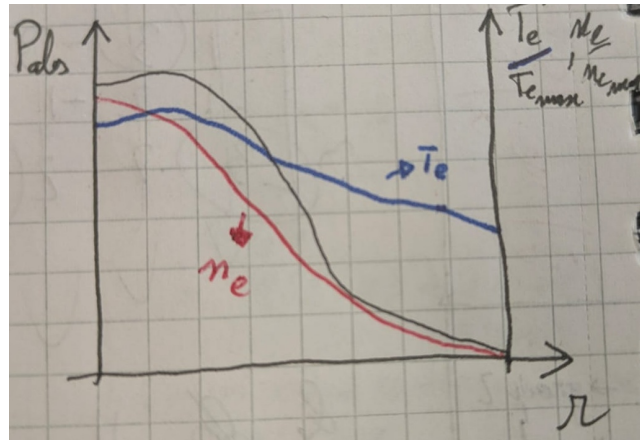


Figure 8: Radial profile for absorbed power density, electron density and electron temperature

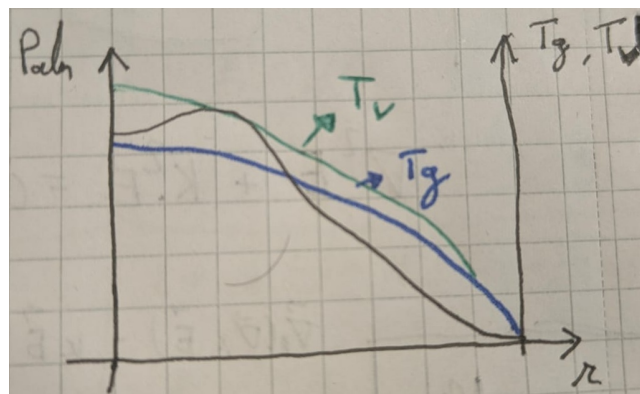


Figure 9: Radial profile for absorbed power density, gas temperature and vibrational temperature

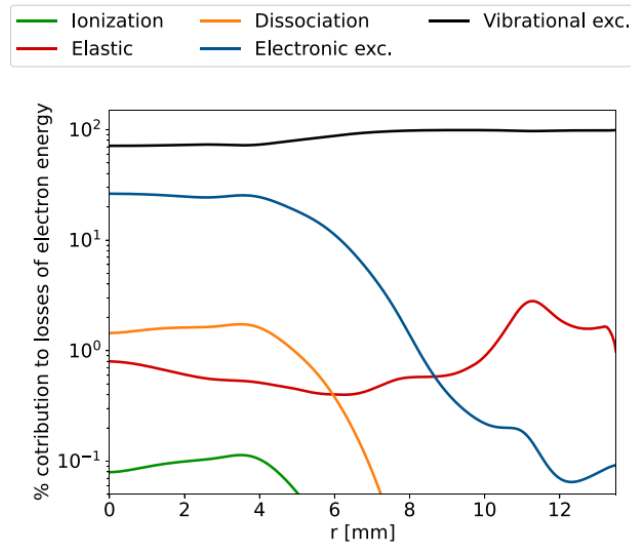


Figure 10: Radial profile of the relative contribution of different processes to the losses of electron energy. (figure from Altin2024, here will be self-consistent power version)

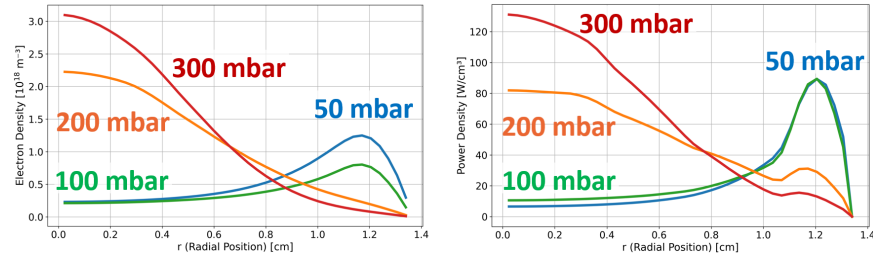


Figure 11: Radial profiles of electron density (left) and absorbed power density (right) at input power 600 W for pressures of 50, 100, 200, and 300 mbar. (maybe also include gas temperature)

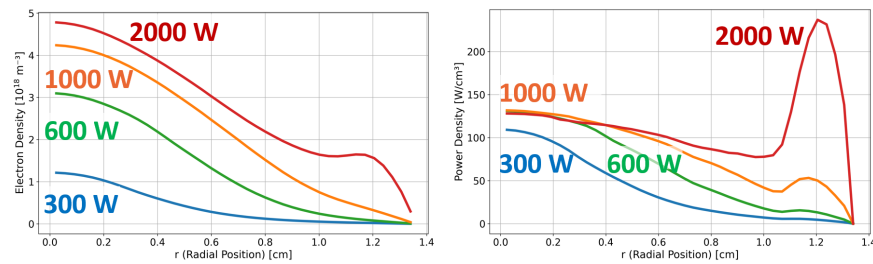


Figure 12: Radial profiles of electron density (left) and absorbed power density (right) at pressure 300 mbar for input powers 300 W, 600 W, 1000 W and 2000 W. (maybe also include gas temperature)

Possible other graphs:

- Radial profile for reaction rates;
- Comparing self consistent runs with WKB, E const and non self consistent runs

4.4 Microwave absorption regimes

Scatter plots describing how the plasma influences the field.

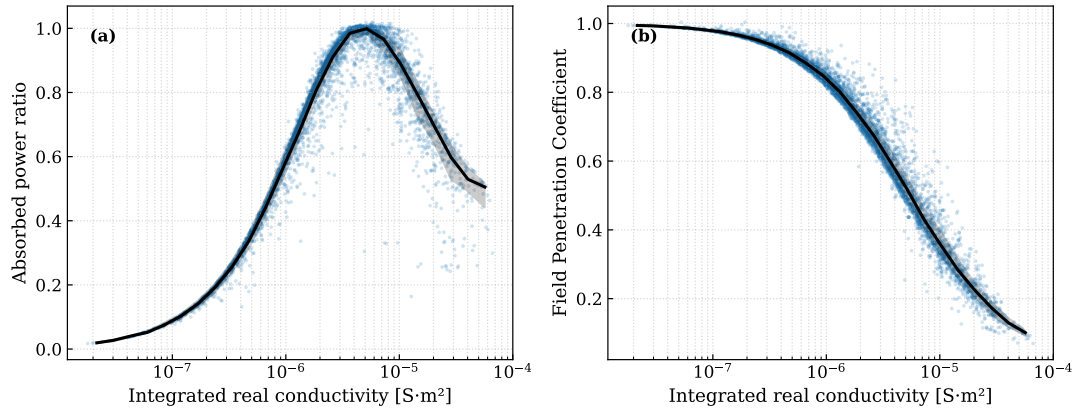


Figure 13: Dependence of (a) the absorbed power ratio and (b) the field penetration coefficient on the integrated real conductivity. Blue markers show individual simulation sampled from a wide distribution of conductivity profiles, while the solid black line represents the median trend computed over logarithmically spaced conductivity bins. The shaded region indicates the 25th–75th percentiles range.

4.5 Electric Field Models Comparison

This section compares the electric field and the power density radial profiles for static plasmas coming from different electric field models:

- Full 2D Maxwell solver;
- WKB approximation for skin depth;
- Constant electric field.

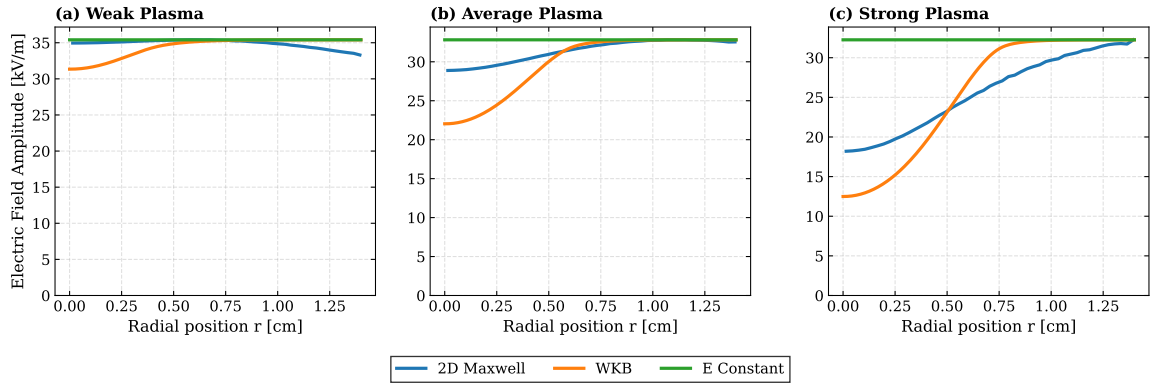


Figure 14: Radial profiles of the electric field amplitude for three plasma conductivity profiles: (a) weak plasma ($n_{e,0} = 8 \times 10^{17} \text{ m}^{-3}$), (b) average plasma ($n_{e,0} = 2.5 \times 10^{18} \text{ m}^{-3}$), and (c) strong plasma ($n_{e,0} = 8 \times 10^{18} \text{ m}^{-3}$). Mobility is kept as $\hat{\mu} = 0.8 - 2.5 \text{ m}^2/(\text{V s})$ in all profiles. Results obtained from the 2D Maxwell solver are compared with the WKB approximation and a spatially uniform electric field model. Approximations are normalized to the maximum values in the 2D Maxwell solution.

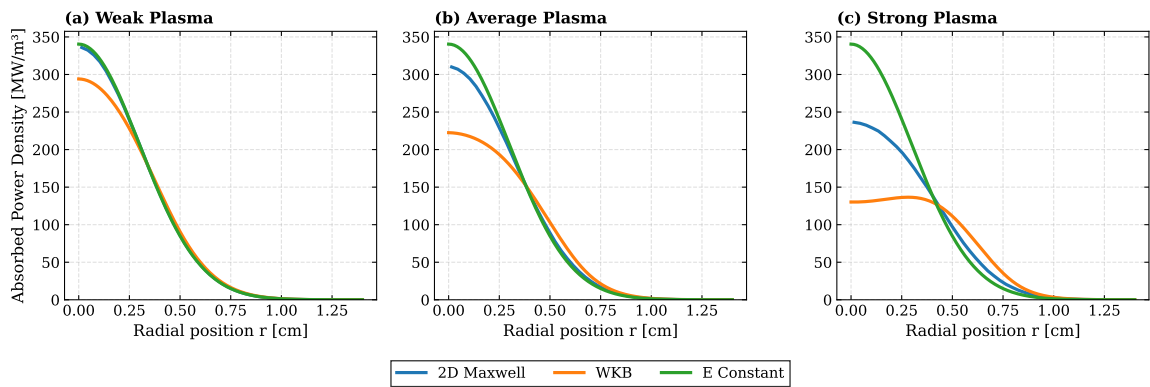


Figure 15: Radial profiles of the absorbed power density for three plasma conductivity profiles: (a) weak plasma ($n_{e,0} = 8 \times 10^{17} \text{ m}^{-3}$), (b) average plasma ($n_{e,0} = 2.5 \times 10^{18} \text{ m}^{-3}$), and (c) strong plasma ($n_{e,0} = 8 \times 10^{18} \text{ m}^{-3}$). Mobility is kept as $\hat{\mu} = 0.8 - 2.5 \text{ m}^2/(\text{V s})$ in all profiles. Results obtained from the 2D Maxwell solver are compared with the WKB approximation and a spatially uniform electric field model. All results are normalized to 770 W of total power.

5 Conclusion

Just a list of short sentences that add a new and higher level of analysis

- Different power density profiles have a significant impact on both internal plasma parameters and macroscopic output quantities.
- Coupling the power density to the fluid model reveals trends with pressure and input power not visible when using static power density profiles.
- Higher input power and lower pressure broaden the power deposition profile, modifying both peak location and penetration depth.
- Impedance matching, even if imperfect, is essential to obtain accurate power density profiles;
- Microwave power coupling with the plasma can be divided into different regimes according to the real part of integrated plasma conductivity;
- Plasmas at very low integrated conductivities cannot have their impedance matched with the microwave source as confirmed in 3D simulations (these simulations show impedance matching is possible up to X).
- Uniform field is only valid at the very lower limits of conductivity, while WKB approximation can be a reasonable approach in cylindrical geometry with the TE_{10} mode for intermediate ranges of integrated plasma conductivity (possibly related these limits to other parameters like power and pressure);
- The present framework provides a foundation for studying different geometries of microwave plasmas and coupling with 3D fluid models.

Data availability

All data that support the findings of this study are included within the article (and any supplementary files).

Acknowledgments

Not sure if I have any funding to declare here.

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